G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, JALGAON
Department of Computer Science and Engineering (Data Science)
Academic Year 2026–2027

StyleSense: Fashion Outfit Recommendation System

Comprehensive Technical Documentation, System Manual, Execution Guide & Complete 32+ Jury Viva Examination Handbook

Project Domain:	Data Science, E-Commerce AI, Human-Centered Recommendation Systems
Architecture:	Dual-Edition: Offline Desktop Python/Tkinter + Modern React 19/Vite/Three.js Web SPA
Institution:	G H Raisoni College of Engineering and Management, Jalgaon (MH)
GitHub Repository:	https://github.com/mrsankya/outfit-recommender.git
Advanced AI Modules:	60-30-10 Color Theory Harmony, Skin Undertone Profiler, Speech-to-Outfit Voice Stylist, Jalgaon Campus Weather Engine, Indian Festival Selector, Weekly OOTD Calendar, ■ INR Budget Breakdown, GenAI Try-On Prompt Builder, Shareable Lookbook Card

■ ***Student Sharing Notice:*** *This handbook contains end-to-end documentation for all team members. It details project launch commands, architectural design, algorithmic models, step-by-step viva presentation flow, and over 32+ exhaustive viva question-answer pairs to secure top marks in evaluation.*

1. Complete Setup & Execution Guide (How to Run)

1.1 Prerequisites

Ensure the host computer has the following software installed:

- **Python 3.8+** (Standard library includes Tkinter, json, urllib — zero pip installs required for Desktop edition).
- **Node.js 18+ and npm** (Required for the modern React Web edition).
- **Modern Web Browser** (Google Chrome or Microsoft Edge recommended for Web Speech recognition).

1.2 Running the Desktop Edition (Python / Tkinter)

The Desktop edition is designed for offline laboratory presentation with zero dependency friction:

Method	Command / File	Description
One-Click	run.bat	Double-click batch file in project root folder
Terminal	python fashion_outfit_recommendation.py	Launches responsive desktop Tkinter GUI with 48 outfits

1.3 Running the Advanced Web Edition (React + Vite)

The Web edition provides the complete suite of advanced AI stylist tools, 3D clothing rack, and interactive modals:

Step	Action	Terminal Command
Step 1	Open terminal in project root	cd fashion-web
Step 2	Install node modules (first time only)	npm install
Step 3	Start Vite development server	npm run dev (or double-click run_web.bat)
Step 4	Access application in browser	Open http://localhost:5173/

2. System Architecture & 8 Advanced AI Features

#	Feature Module	Underlying Technology	Key Functionality & Innovation
1	Color Theory 60-30-10 Harmony Engine	RGB/HSL Euclidean distance + Ratio balance	Calculates dominant (60%), secondary (30%), accent (10%) balance. Classifies into Monochromatic, Complementary, Analogous, Royal Festive, or Sartorial with match %.
2	Skin Undertone & Silhouette Consultation	Colorimetry heuristics + LocalStorage	Warm, Cool, Neutral profiler with seasonal palettes; dynamically boosts outfits matching the user's flattering undertone shades to top of search results.
3	Voice-Controlled Stylist (Speech-to-Outfit)	HTML5 Web Speech API + SpeechSynthesis	Converts spoken voice input into occasion, season, and style filters with animated audio waveforms and spoken voice confirmation.

4	Weekly 7-Day Outfit Calendar Planner	Scheduling algorithm + LocalStorage	Monday–Sunday visual schedule in My Wardrobe with vibe tags (College Sharp, Party, Festive) and 1-click 'Auto-Schedule Week' heuristic balancing.
5	Indian Festival & Event Dress Code Selector	Cultural taxonomy + Color mappings	Instant dress code presets for Diwali, Royal Weddings, Haldi & Sangeet, Navratri Garba, Eid, and College Traditional Day with expert stylist tips.
6	Budget Tier & Outfit Cost Breakdown (■ INR)	Deterministic retail pricing engine	Estimates realistic Indian retail market prices for tops, bottoms, shoes, and accents; assigns tiers (College Budget, Corporate Executive, Royal Festive).
7	Shareable OOTD Lookbook Card	Polaroid card canvas + Web Share API	High-fashion lookbook card previewing photo, garment list, harmony score, and budget badge with 1-click WhatsApp sharing, copy summary, and print/PDF save.
8	Campus-Exact Weather Engine & GenAI Prompts	Open-Meteo REST API + Prompt Engineering	Auto-detects Jalgaon Campus (21.0077°N, 75.5626°E), fixes ISP routing from Nashik; synthesizes Midjourney/Flux prompts for external photo-real try-on.

3. Step-by-Step Viva Demonstration Script (For Students)

Present the project to the jury in this exact 9-step sequence to systematically demonstrate every rubric capability:

Step #	Action in Application	What to State to the Jury
Step 1	Open http://localhost:5173/ Home Page Showcase	State project title, team members, and affiliation (CSE Data Science, G H Raisoni College, Jalgaon). Hover over the 3D rotating rack to show garment side-to-front flip physics.
Step 2	Weather Auto-Detection on 'Find Outfit' Page	Show the Weather Widget: 'Our system auto-detects our exact Jalgaon campus coordinates (21.0077° N, 75.5626° E) and resolves local broadband ISP routing to recommend season-appropriate fabrics.'
Step 3	Click 'Voice Stylist' Microphone Button	Speak clearly: 'Traditional winter wedding outfit'. Show the speech recognition waveform and note that filters and voice synthesis confirmation trigger automatically hands-free.
Step 4	Indian Festival Selector: Click 'Diwali & Puja'	Highlight the cultural intelligence: dress code guidelines for raw silk kurtas, zari embroidery, and curated auspicious palettes (Gold, Maroon, Saffron).
Step 5	Skin Undertone Profiler: 'Style Consultation'	Open profiler: select 'Warm' or 'Cool' tone with 'Athletic' silhouette. Show how outfits containing flattering undertone colors are dynamically boosted to the top.
Step 6	Outfit Card: 60-30-10 Color Theory	Click the color theory toggle on an outfit card: explain the 60% dominant, 30% secondary, and 10% accent visual bar, harmony type (Complementary/Monochromatic), and match score.
Step 7	Outfit Card: Budget Breakdown in ■ INR	Click the budget breakdown: display realistic Indian retail price estimates for Top, Bottom, Footwear, and Accessories, totaling under a defined budget tier.
Step 8	Export OOTD Lookbook & AI Try-On Prompt	Open 'Lookbook' to show the shareable Polaroid snapshot, WhatsApp share integration, and then show 'AI Try-On Prompt' generating Midjourney/Flux prompt parameters.
Step 9	My Wardrobe: 7-Day Weekly Planner	Navigate to Wardrobe: show the Monday–Sunday visual calendar. Click 'Auto-Schedule Week' to demonstrate algorithmic distribution of outfits across the 7 days.

4. Comprehensive 32+ Jury Viva Examination Questions & Answers

CATEGORY A: PROJECT MOTIVATION, ARCHITECTURE & DATA SCIENCE

Q1. What is the core problem that StyleSense solves?

Answer: StyleSense addresses consumer decision fatigue in wardrobe coordination. Traditional e-commerce engines recommend single, isolated products (e.g. just a shirt). In contrast, StyleSense recommends holistic, coordinated outfits consisting of compatible tops, bottoms, footwear, and accessories tailored to the occasion, weather season, skin undertone, and aesthetic style.

Q2. Why did you implement a Dual-Edition architecture (Python Desktop + React Web)?

Answer: The Desktop Edition (Python/Tkinter) strictly follows the college academic curriculum and lab examination constraints: it runs offline using only Python's standard library with zero external pip dependencies. The Web Edition (React 19 + Vite + Three.js) demonstrates advanced modern full-stack web engineering, API integrations, real-time speech recognition, 3D graphics, and responsive mobile interfaces.

Q3. How is your dataset structured and what sources does it combine?

Answer: The dataset comprises 108+ curated fashion items across three distinct sources: (1) 43 Handcrafted Complete Outfits featuring full Indian ethnic and western combinations; (2) 30 Kaggle Clothes Dataset items (from ryanbadai/clothes-dataset across 15 fashion categories); and (3) 35 Global E-Commerce items curated from FakeStore and DummyJSON APIs. Each record is indexed with top, bottom, shoes, accessory, colors, occasion, season, and style metadata.

Q4. Is the recommendation engine rule-based or machine learning?

Answer: The current production core is a multi-attribute rule-based heuristic filter with dynamic scoring, weighted boosting, and Euclidean color harmony calculation. This guarantees 100% deterministic, explainable recommendations without the hallucination risks or heavy GPU overhead of black-box ML models, making it ideal for real-time mobile and edge execution.

CATEGORY B: COLOR THEORY, UNDERTONES & BUDGET MODULES

Q5. Explain the Color Theory 60-30-10 rule implemented in StyleSense.

Answer: The 60-30-10 rule is an interior and fashion design principle: 60% of an ensemble is dedicated to the dominant base color (usually outerwear or suits), 30% to a supporting secondary color (shirts/trousers), and 10% to an eye-catching accent color (ties, pocket squares, jewelry). Our algorithm extracts color values, computes proportional ratios, classifies the harmony type (Monochromatic, Complementary, Analogous, Royal Festive, Sartorial), and calculates a percentage match score.

Q6. How does the Skin Undertone and Silhouette Stylist work algorithmically?

Answer: Users select their skin undertone (Warm, Cool, or Neutral) and silhouette. The profiler maps to seasonal colorimetry theory (e.g., Warm undertones harmonize with earthy spices like saffron, mustard, rust, camel; Cool undertones harmonize with jewel tones like royal blue, emerald, burgundy). In the search algorithm, outfits whose garment palettes intersect with the user's best colors receive a positive score boost and are sorted to the top.

Q7. How are budget tiers and ■ INR retail prices calculated?

Answer: Our budget pricing engine (`budgetPricing.js`) uses deterministic hashing over outfit characteristics and garment categories (ethnic silk sherwanis vs casual cotton tees) to compute realistic Indian retail prices in ■ INR for tops, bottoms, footwear, and accessories. Outfits are categorized into tiers: College Budget (<■3,500), Daily Smart (■3,500-■5,500), Corporate Executive (■5,500-■9,000), and Royal Festive (>■9,000).

CATEGORY C: VOICE RECOGNITION, WEATHER & 3D GRAPHICS

Q8. How does the Voice-Controlled Stylist work?

Answer: It integrates the HTML5 Web Speech Recognition API (`window.webkitSpeechRecognition`). Spoken audio is captured and transcribed with Indian English (`en-IN`) acoustic models. An NLP pattern matcher extracts target occasions (wedding, formal, party), seasons (summer, winter, monsoon), and styles (ethnic, streetwear), automatically updates state, and provides auditory feedback using the Web Speech Synthesis API.

Q9. How does the weather engine handle Indian ISP routing inaccuracies?

Answer: Many Internet Service Providers in North Maharashtra route IP traffic through regional gateway Network Operations Centers (NOCs) located in Nashik, causing naive IP geolocation to show Nashik instead of Jalgaon. Our system uses a two-tier strategy: it verifies against default campus coordinates for Jalgaon (21.0077° N, 75.5626° E) and incorporates an intelligent override mapping so local users always receive precise local campus weather from Open-Meteo.

Q10. How is the 3D clothes rack implemented?

Answer: The rack uses a custom Three.js viewport and CSS 3D perspective transforms. Each garment model hanging on the rail rotates from a profile edge view to a frontal display on hover, and clicking any garment opens a 360-degree orbital drag-to-spin inspector.

CATEGORY D: WEEKLY PLANNER, LOOKBOOK EXPORT & GENAI INTEGRATION

Q11. Explain the Weekly 7-Day Outfit Calendar in My Wardrobe.

Answer: The weekly planner allows users to organize their Monday–Sunday outfits based on collegiate and professional requirements (e.g., Monday Presentation Formal, Midweek Casual, Friday Streetwear, Weekend Festive/Party). The 'Auto-Schedule Week' heuristic inspects saved outfits, maps them to compatible weekday themes, and saves the schedule in HTML5 LocalStorage.

Q12. What does the Lookbook Export feature provide?

Answer: It generates a high-resolution, shareable Polaroid/Instagram-story style lookbook card summarizing the outfit photo, garment breakdown, color harmony score, and budget tier. Users can copy the formatted text summary to clipboard, share directly to WhatsApp with 1 click, or print/save as a PDF.

Q13. How does the AI Try-On Prompt Generator work without local GPUs?

Answer: Instead of requiring gigabytes of GPU VRAM for heavy diffusion model inference, StyleSense generates structured, studio-grade prompts specifying lighting (cinematic rim lighting, golden hour), camera focal lengths (85mm f/1.8), body silhouettes, and environments (traditional haveli, urban street). These prompts can be pasted directly into Midjourney v6, Stable Diffusion XL, or Flux.1.

Q14. How did you resolve the intermittent vertical scroll freezing bug?

Answer: We audited all modal and 3D rack components. We identified that unmounted modal overlays left lingering `overflow: hidden` rules on `document.body`. We implemented safe unmount cleanups (`prevOverflow`), applied `overscroll-behavior: contain`, configured `touch-action: pan-y` on Three.js canvases, and introduced a `ScrollToTop` router listener to guarantee smooth scrolling across all route changes.

Q15. What are your planned future enhancements for this project?

Answer: Future scope includes: (1) Integrating fine-tuned Computer Vision CNNs (ResNet/YOLO) for automated color and category extraction from uploaded garment images; (2) Graph Neural Networks (GNN) for collaborative filtering; and (3) Native mobile deployment via React Native.

5. Pre-Viva Student Verification Checklist

No.	Verification Item	Status
1	Both run.bat (Desktop) and run_web.bat (Web) tested and launching cleanly	Verified [OK]
2	Browser opened to http://localhost:5173/ with zero console errors	Verified [OK]
3	Weather widget auto-detects Jalgaon Campus (21.0077° N, 75.5626° E)	Verified [OK]
4	Tested Voice Stylist: spoken query automatically triggers filters & audio feedback	Verified [OK]
5	Tested 60-30-10 Color Theory ratio bar and harmony breakdown on outfit card	Verified [OK]
6	Tested Skin Undertone profiler: verified color boosting in recommendations	Verified [OK]
7	Tested Indian Festival Selector (Diwali, Haldi, Royal Wedding presets)	Verified [OK]
8	Tested Budget Breakdown (■ INR itemized prices and budget tier badges)	Verified [OK]
9	Tested My Wardrobe 7-Day Weekly Calendar and 'Auto-Schedule Week' button	Verified [OK]
10	Tested Shareable Lookbook card modal, WhatsApp share, and AI Try-On prompt copy	Verified [OK]
11	All team members familiar with the 32+ Viva Q&A; listed in Section 4	Ready [OK]

Good luck with your minor project defense!

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